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import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, confusion_matrix

import seaborn as sns

import matplotlib.pyplot as plt


# Sample email dataset

data = {
    "email": [
        "Congratulations! You won a free prize. Click this link now.",
        "Your account has been blocked. Verify your password immediately.",
        "You have won a lottery. Send your bank details to claim the prize.",
        "Urgent! Click the link to verify your account.",
        "Your password will expire. Login now to update it.",
        "Claim your free gift by clicking this URL.",
        "Meeting is scheduled for tomorrow at 10 AM.",
        "Please find the project report attached.",
        "Your college class starts at 9 AM tomorrow.",
        "Can you send me the assignment file?",
        "The team meeting has been moved to Friday.",
        "Thank you for submitting your project."
    ],
    "label": [
        "Phishing", "Phishing", "Phishing",
        "Phishing", "Phishing", "Phishing",
        "Safe", "Safe", "Safe",
    ]
}
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        "Safe", "Safe", "Safe"
    ]
}

# Convert data into DataFrame
df = pd.DataFrame(data)

# Split data into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    df["email"], df["label"], test_size=0.25, random_state=42
)

# Convert text into numbers
vectorizer = TfidfVectorizer()
X_train = vectorizer.fit_transform(X_train)
X_test = vectorizer.transform(X_test)

# Train the machine learning model
model = LogisticRegression()
model.fit(X_train, y_train)

# Test the model
prediction = model.predict(X_test)

# Display accuracy
accuracy = accuracy_score(y_test, prediction)
print("Accuracy:", accuracy * 100, "%")
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# Confusion Matrix

cm = confusion_matrix(y_test, prediction)

sns.heatmap(

    cm,

    annot=True,

    fmt="d",

    xticklabels=["Phishing", "Safe"],

    yticklabels=["Phishing", "Safe"]

)

plt.xlabel("Predicted")

plt.ylabel("Actual")

plt.title("Phishing Email Confusion Matrix")

plt.show()

# Test your own email

email = input("\nEnter an email to check: ")

email_vector = vectorizer.transform([email])

result = model.predict(email_vector)

print("Result:", result[0])
```

RESULTS AND PROOFS:

```
$ python phishing_detector.py
Accuracy: 33.33333333333333 %

Enter an email to check: Please send me the project report tomorrow.
Result: Safe
```

```
$ python phishing_detector.py
Accuracy: 33.33333333333333 %

Enter an email to check: Congratulations! You won a free prize. Click this link now.
Result: Phishing
```

```
$ python phishing_detector.py
Accuracy: 33.33333333333333 %

Enter an email to check: The team meeting has been moved to Friday.
Result: Phishing
```

